

VIDEO 1

QUARTILE DEVIATION AND IQR



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Probability And Statistics

Quartile Deviation and IQR

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Introduction



In this video we will discuss

- ▶ Quartile deviation
- ▶ Inter Quartile Range



Quartile Deviation

Quartile Deviation is a measure of dispersion in statistics.

It is also called Semi-Interquartile Range.

It shows how data values are spread around the middle 50% of a dataset.



Quartile Deviation

Quartile Deviation is defined as half of the difference between the third quartile (Q_3) and the first quartile (Q_1).

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

where Q_1 = First Quartile (25th percentile)

Q_3 = Third Quartile (75th percentile)



Inter Quartile Range(IQR)

Inter Quartile Range(IQR) is defined as the difference between the third quartile (Q_3) and the first quartile (Q_1).

$$IQR = Q_3 - Q_1$$

where Q_1 = First Quartile (25th percentile)

Q_3 = Third Quartile (75th percentile)



Example

Find IQR of the following Data:

11,21,18,13,12,16,16,17,17

Data in Ordered Array:

11 12 13 16 16 17 17 18 21

$$\text{Position of } Q_1 = \frac{1(9+1)}{4} = 2.50$$

$$Q_1 = \frac{12+13}{2} = 12.5$$

$$\text{Position of } Q_3 = \frac{3(9+1)}{4} = 7.50$$

$$Q_3 = \frac{17+18}{2} = 17.5$$

$$\text{Interquartile Range} = IQR = Q_3 - Q_1 = 17.5 - 12.5 = 5$$

